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Smart Home Based Ambient Assisted Living

Recognition of Anomaly in the Activity of Daily Living for An Elderly Living Alone

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Abstract—**Many people with incapacities can continue lives in their existing home with the application of an assistive informatics and healthcare monitoring. Elderly's ability to perform daily activities presents the elderly's wellness. Investigation of human activities for extracting health and well-being statistics is one of the present challenges given the Smart Aging. The existing research works are mostly focusing on offline monitoring and do not give significance to real-time monitoring and forecasting. This paper aims to apply lifestyle monitoring and forecasting in real time in ambient assisted living (AAL) system with diagnosing the conduct and distinguishing the variation from the norm with the slightest conceivable fake alert. Anomaly forecasting model based on inactivity and activity of period of the different ADL is formulated. The present study demonstrates the novel approach for forecasting of wellness anomaly related to the activity of daily living (ADL). The significant correlation between lifestyle and health status is established. Through the present research forecasting of lifestyle would facilitate remote physician/caregiver to insight symptom of the disease and provide health improvement advises to the resident. Moreover, if it is a life-threatening degradation in wellness index than appropriate service provider would be informed directly.**

Keywords—AAL, ADL, anomaly, forecasting, pattern generation.

I. RESEARCH CHALLENGES AND NECESSITY

Smart home technology has expanded from daily usage objects to healthcare at home. The objective of ambient assistant scenario incorporates smart home technology to live independently. Sensor activation based smart home monitoring keeps caregiver updated about the inhabitant [1-3]. A usual concern for the caregiver is the ability to immediately alert and be available in case of any life threating condition or bad is going to occur to elderly. For this issue, the smart home system does monitoring, analysis and ultimately forecasting to predict the bad day for a participant. To forecast the aspects of an individual's day, designing of intelligent computational approaches requires [4-6].

This paper achieves this goal by measuring the activity of daily living demonstrated by usage of the different household object, electrical and electronic appliances and movement inside the home based on sensor activation [9]. The Smart Aging sensor data fusion can be classified into two domains; one part is activity discovery and behavioral pattern generation that has been explored in last journal articles [7-9]. The AAL conditions demand the event detection of the ADL from the raw sensor activation, and these sensory datasets are puzzling and irregular to translate into predefined situations. Without a doubt, even in the endeavor of encoding this raw data, it is difficult to pledge the degenerative conduct because these dataset's informational indexes are on the distinctive time measures (sampling rate) and sensing modalities. This diverse testing rate and sensing modalities grounds troublesomeness during the condition (event) definition.

To give an accurate methodology, we proposed and completed an AAL in view of an integrated framework to explore the individual's routine from the past information, ongoing as of late got information and acknowledged got information. To address the research challenges, we apply behavioral discovery through the pipeline approach with pre-preparing, movement acknowledgment, and smoothing. We center to separate the general routine data from astounding data, which may trigger risk to health.

We continue the research work by presenting the forecasting methodology in Section II, with a short description of wellness indices formulation. The experimental results, observations & performance issues in Section III have been represented. The extensions of a present research for future work and a conclusion with conceivable arrangements towards issue determination is attracted Section IV.

II. METHODOLOGY FOR WELLNESS FUNCTION DESIGN

The smart home used diverse wireless sensors, those assorted sensors were temperature sensing unit, force sensing unit, movement monitoring unit (passive infrared sensor, contact sensing unit, smoke sensing unit, Manual Alert button and electronic and electrical (E & E appliances sensing) appliances monitoring unit. The sensors were positioned in such a style so that the system got each indispensable sensor activation from every corner of the AAL environment, which was valid to wellness determination of an elderly inhabitant [7-9]. Fig.1 shows the stepwise implementation of smart home approach and Fig.2, offers the overview of smart home system deployment.

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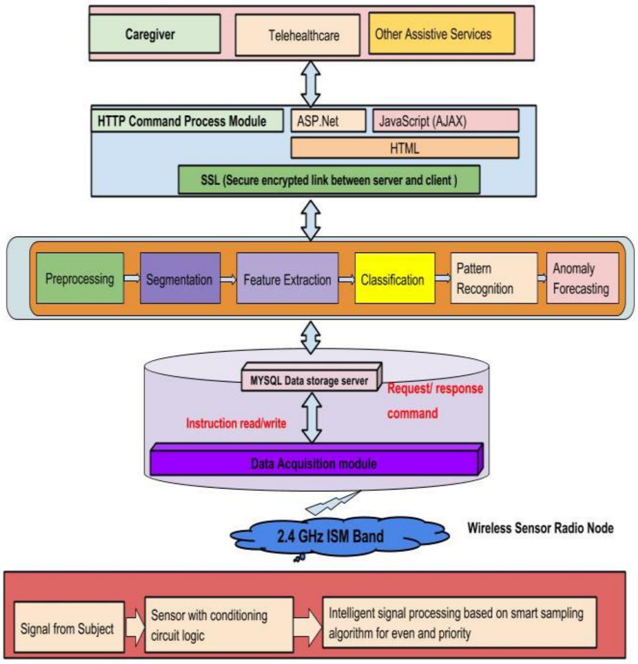
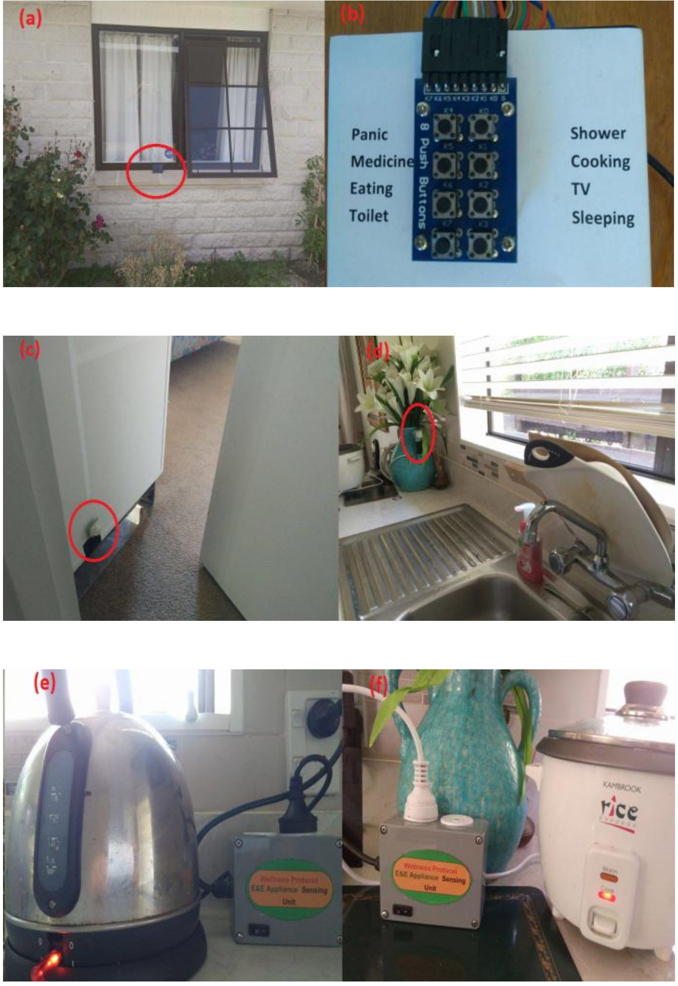

Caregiver
Telehealthcare
Other Assistive Services
HTTP Command Process Module
ASP.Net
JavaScript (AJAX)
HTML
SSL (Secure encrypted link between server and client)
Preprocessing
Segmentation
Feature Extraction
Classification
Pattern Recognition
Anomaly Forecasting
MYSQL Data storage server
Request/ response command
Instruction read/write
Data Acquisition module
2.4 GHz ISM Band
Wireless Sensor Radio Node
Signal from Subject
Sensor with conditioning circuit logic
Intelligent signal processing based on smart sampling algorithm for even and priority

Figure.1: Bottom-up diagram to represent the Smart Home based AAL system



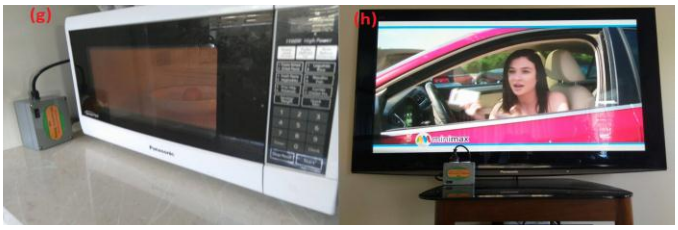


Figure 2. (a): Outdoor temperature sensing unit, (b): Activity-based manual Indication unit, (c): Movement Sensor deployed at the entry, (d): Movement Sensor deployed at the corner of washbasin in kitchen, (e): Water Cattle plugged to E & E Sensor, (f): Rice cooker plugged to E & E Sensor, (g): Microwave Oven plugged to E & E Sensor &, (h): Television plugged to E & E Sensor.

The medicinal services experts and the guardian would help in the event that they get totally extricated information in easy to understand organize rather than the timestamp of exercises or household use, area. The digital representation is finest methods to connote the wellbeing of an occupant. The β1 and β2, are two wellbeing parameters created to quantify the prosperity of an inhabitant. The computation of β1 is finished by the non-utilization or inert interim of the household while β2 is assessed by dynamic utilization interim of the household. On the off chance that β1 and β2 are equivalent to 1, at that point, the elderly tenant is fit and fine. At first, Wellness function β1, old and β2, old were figured. Later, we have modified the [7] our previously designed old wellness functions. The new indices have modified by introducing the different observations of seasonal and random days to minimize the issue of false alarm [7-9]. The worry with past framework was, it did not involve the random/seasonal deviations and that causes false caution messages. In new wellbeing capacity, the occasional changes perceptions have been presented through the time series approach [8]. The present approach is dynamic approach, where the wellness indices

$$\beta_{1,new} = e^{-t/T_{IN}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Where $\beta_{1,\ new}$ wellness function is formulated through the inactive time duration of object usage by the elderly inhabitant; t is the actual time duration of inactivity of all or specific object, when they were not in use; T_{IN} is the maximum allowable inactive time duration during which no object was used and may possibly cause anomaly condition. This duration is calculated from past observations. In for the regular life of an individual β_1 should be 1.

$$\beta_{2,new} = e^{\frac{(T_N - T_A)}{T_n}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Where β_2, new wellness function is designed through the excessive active time duration of object usage by the individual; T_A is the actual time interval of activity of a particular object when they were in use. T_N is the maximum active time interval during which object was used, and beyond this time duration, it may possibly cause anomaly condition. This duration is calculated from past observations. In for the regular life of an individual β_2 should be 1. The abnormality is only calculated when $T_A > T_N$.

The standard span of object utilization indicates that the elderly tenant is performing regular exercises in a sound way. Presenting to occasional changes, the elderly's conduct and the era of object utilization changes, this is not an oddity condition. It ought not be assigned as a preventative alarm. For an illustration, an elderly's every day exercises are not going to stay same from winter to summer; there will be a considerable change in the exercises. In this manner, the behavioral change should be perceived and isolated from irregularity sign. In light of the ADL acknowledgment, we will estimate the hazard causing occasion later on.

Displaying Trends and Forecasting: The appraisal of expectation machine learning model is done by current state and recorded sensory datasets. Fig.3 presents the learning of gauging model. The sensory datasets are arranged into three sets, training, cross-validation and testing. Training data create the model with the Wellness Learning Algorithm. The moment Wellness Model is created; the past test datasets are provided to evaluate the model accuracy.

To predict an ADL continuously, a progression of stages are characterized as follows: • Step I-Calculate the non-specific wellbeing capacity parameters from recently gathered authentic events, regular information from past months or season and assess health files from the scientific formula (1) and (2).

• Step II-Select one irregular day 24 hours' dataset as the test dataset and apply rest of the proper dataset for the training, change and refresh the model on the test set [7]. Repeat the same for cross-validation.

• Step III-Formulate the ADL display by the use of training dataset [9].

• Step IV-Execute the time series technique through the test set in an ongoing or close constant information spilling; for each inbound tactile occasion initiation, apply the sensory data segmentation strategy and use the prebuilt movement model to figure the new action tag.

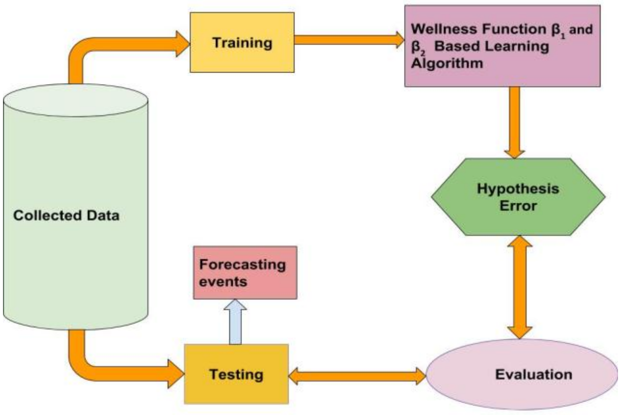


Figure 3: Working block diagram of Wellness Learning Model for time series examination

III. BEHAVIOURAL TREND GENERATION AND FORECASTING

The data from different unobtrusive sensing units had been received in the local home gateway database server for trend generation and anomaly detection. The pattern detection and generation of behavior is computing the most appropriately fitted curve from the trends and seasonal parameters.

Behavioral pattern generation in this methodology, a procedure is introduced for obtaining the elderly behavior and differentiate the anomaly from normal conduct with the least possible fake alert. For the precise evaluation of routine change and anomaly discovery, T_{IN} and T_N time of object usage were framed according to days, weeks, months of the calendar and seasonal changes. Time series machine learning techniques have been used to derive the model for T_{IN} and T_N.

Definition of the symbol used in formulas from equations (3) to (9) [7-9]:

L=number of period in one cycle or seasonal length (for a week 1 to 7)
α= Data smoothing factor 0 to 1
β =Trend smoothing factor 0 to 1
γ= Seasonal change smoothing factor 0 to 1
x= Observation of object at current time
S= Smoothed observation
T_t= Trend factor
C= Seasonal Trend/ Index
F= The forecast at "m" periods ahead
t= Index to show the time period
m= number of steps ahead of forecast

$$F_{t+m}=S_t+ mT_{IN}+ C_{t-L+1+((m-1) \bmod L)} \ldots\ldots\ldots\ldots(3)$$

$$F_{Nt+m}=S_{Nt}+mT_N+C_{Nt-L+1+((m-1) \bmod L)} \ldots\ldots\ldots\ldots(4)$$

$$S_t=\frac{1}{L}\left(x_1+x_2+x_3+x_4 \ldots\ldots x_L\right.\ldots\ldots\ldots\ldots(5)$$

$$S_{Nt}=\frac{1}{L}\left(x_1+x_2+x_3+x_4 \ldots\ldots x_L\right.\ldots\ldots\ldots\ldots(6)$$

$$C_1=x_1-S_t,\ C_2=x_2-S_t,,,,,,,,,,,,\ldots\ldots\ldots\ldots\ldots(7)$$

$$C_{N1}=x_1-S_{Nt},\ C_{N2}=x_2-S_{Nt},,,,,,,,,,,,\ldots\ldots\ldots\ldots\ldots(8)$$

$$T_t=\left[\left(\frac{1}{L}\right)\left\{\left(\frac{x_{L+1}-x_1}{L}\right)+\left(\frac{x_{L+2}-x_2}{L}\right)+\left(\frac{x_{L+3}-x_3}{L}\right)+,,,,,,,,,,,,+\left(\frac{x_{2L}-x_L}{L}\right)\right\}\right]\ldots\ldots\ldots(9)$$

The conduct of the elderly inhabitant was characterized as regular or anomaly depends on the equations derived. For the forecasting, the current activity duration was checked correlated with the forecasted duration. The confidence level of 95% was assumed in the forecasting analysis. The acceptable limit of duration of any routine activity was defined by equation (10) below. If the actual duration was beyond the limit according to equation 10 then an anomaly flag was triggered.

Duration of routine activity = Forecast duration ± 2*standard deviation ……. (10)

Fig.4(A), 5(A) & 6(A) show the activity duration and its equivalent trend analysis for six weeks at an elderly inhabitant's house living alone, whereas Fig.4(B), 5(B) & 6(B) show the activity duration and its equivalent forecasting analysis for one week at an elderly inhabitant's house living alone.

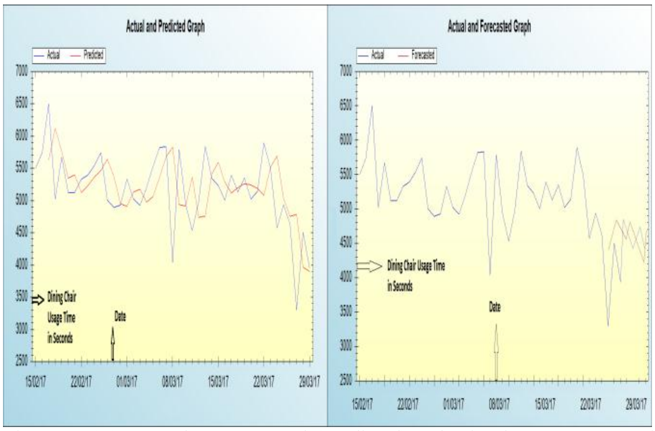


Figure 4: (A). Actual versus Trend (Predicted) Behavioral Pattern Generation for Dining Chair Usage (Eating Activity), demonstrates the eating movement (feasting seat utilization) term and its equivalent trend investigation for a month, (B). Actual versus forecasted (7 days) Behavioral Pattern Generation for Dining Chair Usage (Eating Activity), Portrays the use of eating seat by a tenant for a month and its comparing pattern, prediction for the following week.

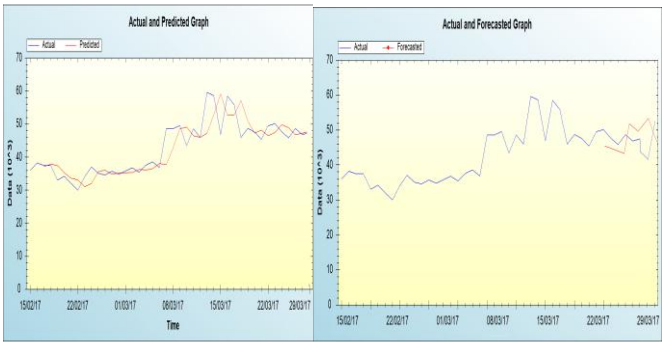


Figure 5: (A). Actual versus Trend (Predicted) Behavioral Pattern Generation for Bed Usage (Sleeping Activity), demonstrates the sleeping action (bed use) term and its comparable trend examination for a month (B). Actual versus forecasted (7 days) Behavioral Pattern Generation for Bed Usage (Sleeping Activity), Delineates the use of bed by a tenant for a month and its equivalent trend, prediction for the following week

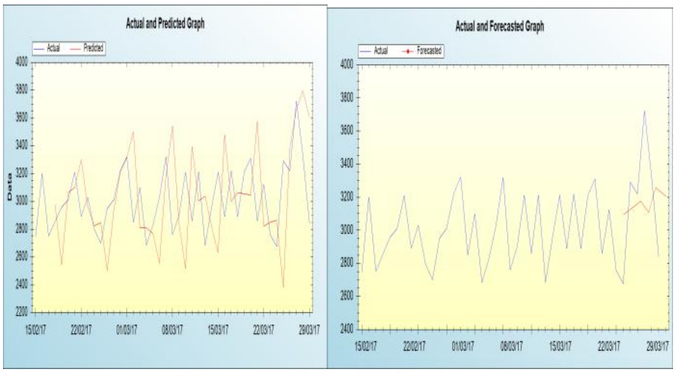


Figure 6: (A). Actual versus Trend (Predicted) Behavioral Pattern Generation for Toilet Seat Usage (Toilet Activity), shows the Toilet activity for six weeks, (B). Actual versus forecasted (7 days) Behavioral Pattern Generation for Toilet Seat Usage (Toilet Activity) for six weeks and its equivalent trend, the forecast for the next week

Based on the formula derived 1-10, the values of T_{IN} and T_N had been measured, which is accessible in Table 1. The Smart Aging system was deployed in four different elderly subject house where elderly was living alone, below data analysis is based on their AAL.

Table 1: Wellness Parameters of household utilization and prediction of the ADLs

S U B	Activity	Sensor_ID	β₁	β₂	Forecasting for an upcoming week					Actual-Duration (Sec)	Status
					Max-Time (Sec)	Min-Time (Sec)	α	δ	Γ		
#1	Sleeping	Bed	0.847	0.826	29583	25703	0.180	0.210	0.398	25900	Regular
	Dining	Chair		0.901	4672	3827	0.140	0.058	0.472	3953	Regular
	Toilet	Toilet		**0.592**	**2014**	**1649**	**0.027**	**0.082**	**0.432**	**2348**	**Anomaly**
	Shower	S-Door		0.894	1389	1039	0.048	0.284	0.472	1309	Regular
	Relax	Couch		0.958	1843	1523	0.194	0.321	0.573	1648	Regular
	Watching TV	TV		0.859	3728	2751	0.064	0.184	0.284	3392	Regular
#2	Sleeping	Bed	0.802	0.841	28494	23784	0.044	0.173	0.583	27490	Regular
	Eating	Chair		**0.507**	**4849**	**3392**	**0.291**	**0.055**	**0.382**	**5534**	**Anomaly**
	Toilet	Toilet		**0.473**	**1931**	**1573**	**0.060**	**0.292**	**0.539**	**2649**	**Anomaly**
	Shower	S-Door		0.793	1502	1237	0.032	0.380	0.284	1392	Regular
	Relax	Couch		0.884	1483	1148	0.200	0.440	0.480	1410	Regular
	Watching TV	TV		**0.382**	**4682**	**3871**	**0.200**	**0.060**	**0.500**	**1639**	**Anomaly**
#3	Sleeping	Bed	0.854	**0.551**	**32280**	**27484**	**0.020**	**0.460**	**0.505**	**37402**	**Anomaly**
	Eating	Chair		**0.403**	**4782**	**3526**	**0.200**	**0.520**	**0.150**	**1274**	**Anomaly**
	Toilet	Toilet		0.850	1493	1174	0.440	0.400	0.500	1300	Regular
	Shower	S-Door		0.930	1648	1382	0.200	0.180	0.400	1532	Regular
	Relax	Couch		0.935	1702	1485	0.300	0.350	0.500	1643	Regular
	Watching TV	TV		0.940	3404	2940	0.300	0.250	0.400	3100	Regular
#4	Sleeping	Bed	0.864	0.912	30291	27493	0.200	0.500	0.350	28490	Regular
	Eating	Chair		0.945	3784	3028	0.300	0.500	0.400	3678	Regular
	Toilet	Toilet		0.834	2630	2134	0.200	0.370	0.650	2219	Normal
	Shower	S-Door		0.874	1835	1480	0.300	0.350	0.600	1743	Regular
	Relax	Couch		0.956	1539	1239	0.200	0.300	0.500	1503	Regular
	Watching TV	TV		0.883	3921	3283	0.060	0.100	0.360	3528	Regular

There is one thumb rule present Smart Aging system is following, whenever anomaly detected in a sub-activity or activity do not alarm, please check that event with multi-sensor data fusion to avoid the false positive. From the observation of the above table records, it is detected the only subject house-4 was free from an anomaly, on the other hand, rest of the subject houses had anomaly forecasting tags. From subject 1 and 2 they had over usage of toilet activities, but both had different causes of an anomaly. In case of subject-1, stomach upset, whereas subject-2 had a visitor at home. Additionally, due to the visitor subject-2 was sitting more on a dinning chair while eating and that caused the anomaly of the chair over usage. Subject-3 was in the unhealthy condition that caused him sleeping a lot and eating less.

To validate the implemented algorithm, the dataset previously described has been considered. There are four possible results:

For ADL

1. True positive (TP): The “A” activity happens and the algorithm detects it correctly the “A” activity .
2. False positive (FP): The “A” activity does not occur and the algorithm reveals the “A” activity.
3. True negative (TN): a daily event is performed and the algorithm does not detect it the “A” activity.
4. False negative (FN): The “A” activity happens but the algorithm does not detect it.

For anomaly

1. True positive (TP): a anomaly happens and the algorithm detects it.
2. False positive (FP): a anomaly does not occur and the algorithm reveals a anomaly.
3. True negative (TN): a daily event is performed, no anomaly occurs and the algorithm does not detect it.
4. False negative (FN): a anomaly occurs but the algorithm does not detect it.

The performance of the system can be evaluated considering the following metrics:

$$\text{Sensitivity}= TP/(TP+FN) \quad (10)$$

$$\text{Specificity}= TN/(TN+FP) \quad (11)$$

$$\text{Accuracy}= (TP+TN)/(TP+FP+TN+FN) \quad (12)$$

The sensitivity, specifity and accuracy for ADL were 96.12%, 96.97% and 96.53% respectively. Whereas the sensitivity, specifity and accuracy for anomaly detection were 92.31%, 95.10% and 93.69%.

IV. Conclusion and Future Work

The basic objective of the human data information examination by time series was to find a model which could figure the authentic characteristics of the arrangement, as the model will empower us to envision next estimations of the series from its recorded data. This is significant to the routine of the elderly individual changes with time and the movements should be thought in the model. The accompanying advances were performed in the investigation of information utilizing the time series strategy: Time arrangement of past data, a suitable method for parameters estimation, the best fitting model for investigation and guaging of new esteems. The estimation of regular activities was done through the usage of household sensor data. Foreseeing the lead of an elderly individual relied upon past sensor activity terms. A Combination of detecting structure with time arrangement tactile information preparing capacity will empower us to measure how well an elderly individual can execute their regular routine consistently. Up until this point, the anticipating method could appropriately check the prosperity records related to the use of non-electrical appliances. Subsequently, a portion of the fundamental elderly individual's day by day exercises, for example, sleeping, toileting, eating and relaxing were appropriately evaluated by the wellness estimation framework. Since the majority of the electrical apparatuses use terms (duration) are predefined; validation for exercises, for example, preparing food was restricted. Forecasting the usage term of household objects in the keen home by time series model can correctly anticipate to what extent a specific appliance can be utilized by the elderly individual. This model was useful in diminishing the quantity of false alerts to be produced. This will empower us to have supplementary data in the information investigation for successful longitudinal gauging of the detecting terms.

The system needs to redefined with some more parameters and observation because presently it needs more than 80% of training data for the low false positive to anomaly detection.

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